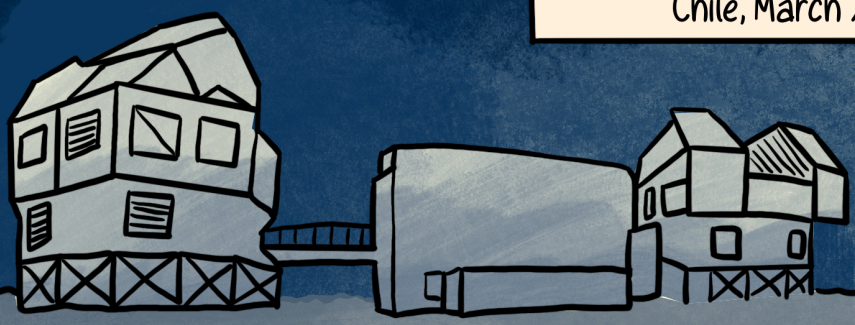
Enriched by Previous Generations

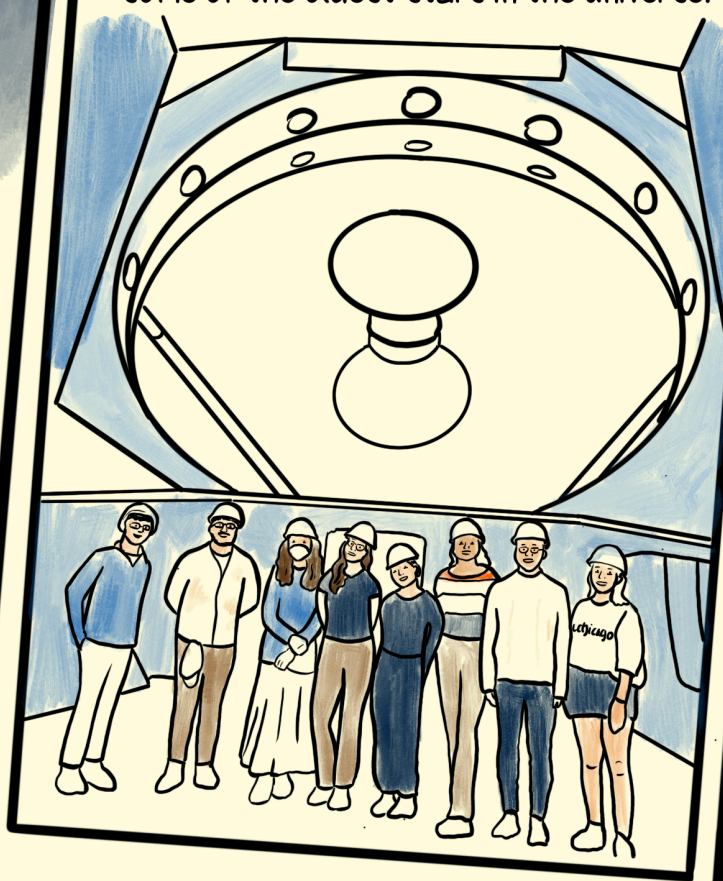
How a team of undergraduates discovered a child of the first stars

By Elijah Tan

Las Campanas Observatory (LCO)
Chile, March 2025



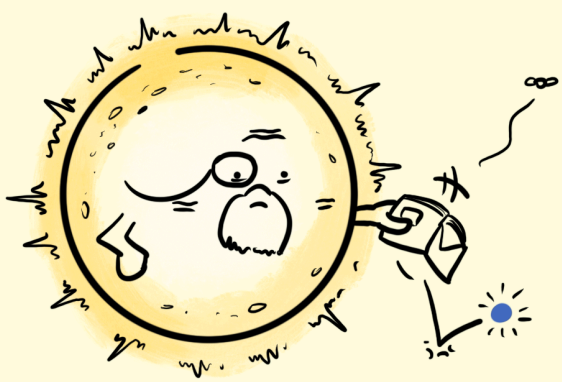
A team of undergraduates from the University of Chicago were scouting for some of the oldest stars in the universe.



The team with LCO's Clay Telescope

Their targets:
A shortlist of *metal-poor stars*.

These stars tend to be older,



While younger stars are more likely to be *metal-rich*.

Hydrogen

Helium

In astronomy, the term "metals" means something quite different from its everyday use.

Every element that is not hydrogen or helium is called a "metal".

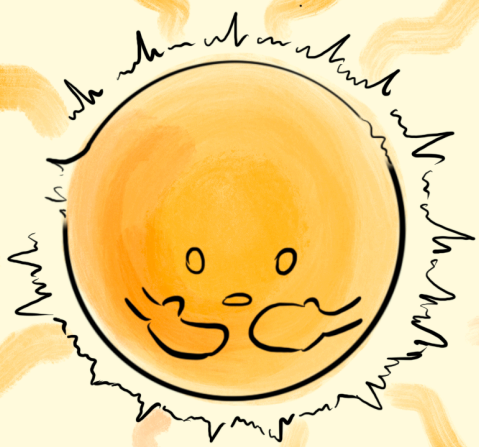
Li Be
Na Mg
K Ca Sc Ti V Cr Mn Fe Co Ni Cu Zn Ga Ge As Se Br Kr
B C N O F Ne
Al Si P S Cl Ar

Metals

In the early universe, only hydrogen and helium existed,

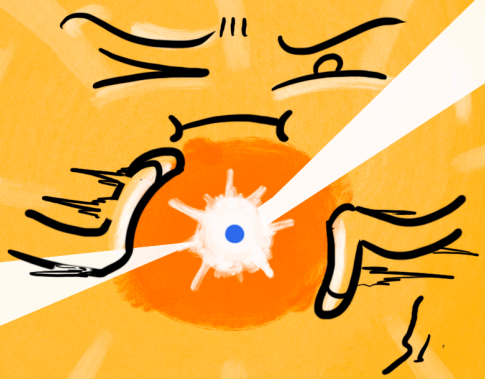
Filling the cosmos with their pristine gas.

From this pristine gas came the very first stars.



Population III Stars, the oldest and metal-poorest of them all.

In their cores, they began to forge the first metals through nuclear fusion.



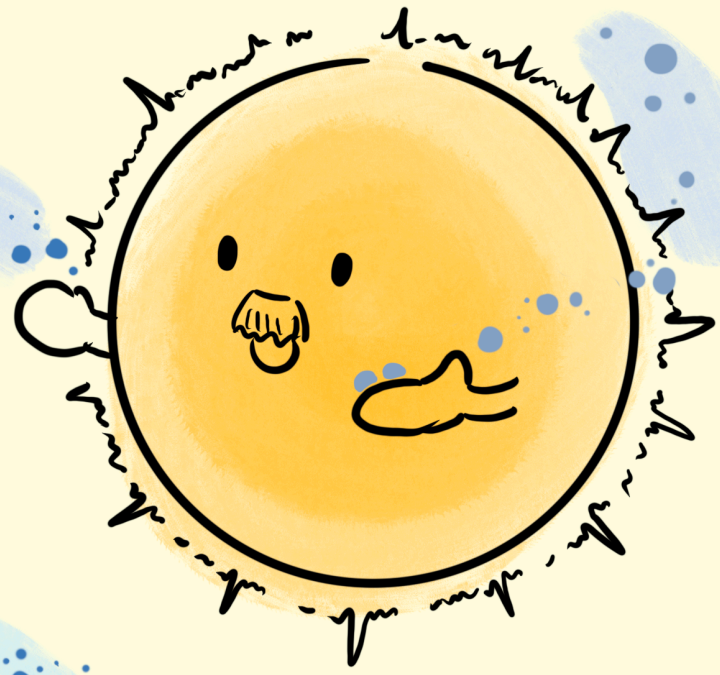
Crushing hydrogen and helium into heavier elements to emit their heat and light.

Till they died in enormous explosions known as **supernovae,**

Creating even more complex metals from the sudden, enormous pressure and heat.

Their matter, and all of their newly created metals, were released back into the universe...

To form the next generation of stars.



With every supernova that followed, old stars passed down a growing inheritance of metals to new stars,

And even new planets,

Which in at least one documented case, made possible the formation of life.

You can estimate how much material was made in the explosions of the first stars, and how much of that is in our bodies today.

Professor Alex Ji, University of Chicago

It's about this much.

Half a pinky finger, or about a teaspoon of you, is made in the very first generation of stars.

Ji led the undergraduates to Chile, as the instructor of the students' Field Course in Astronomy.

Why are you interested in Population III stars?

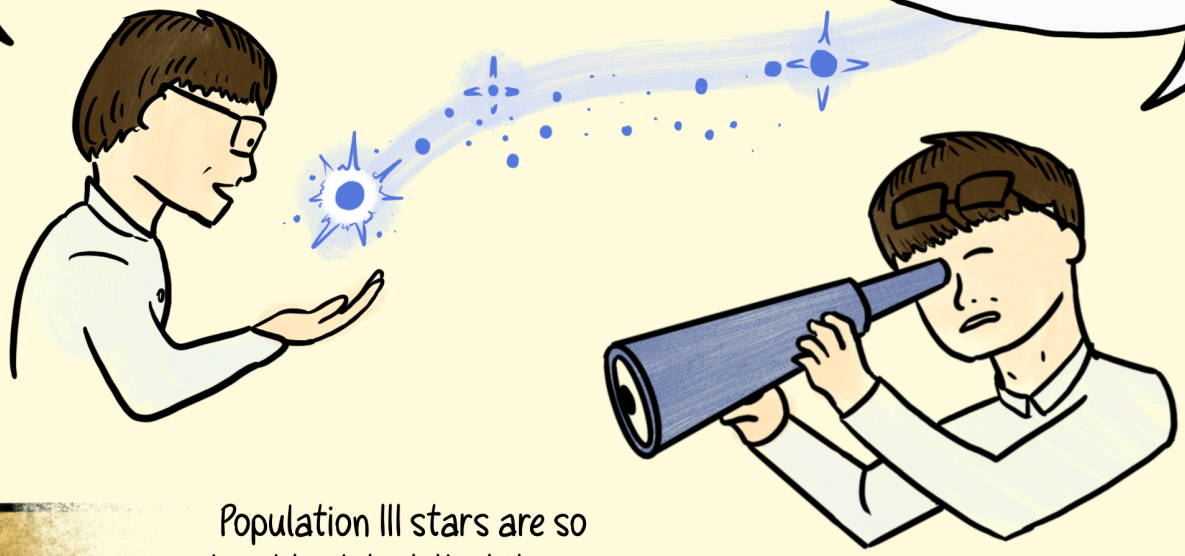
I'll be perfectly honest.

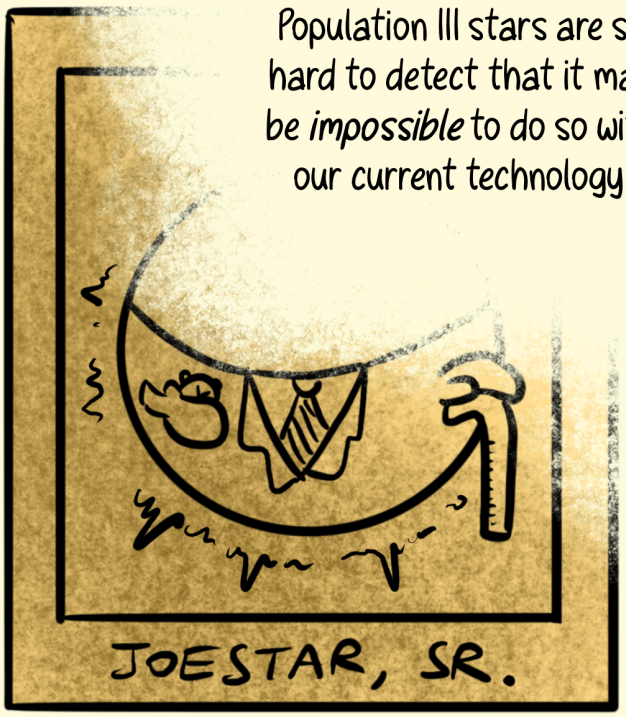
I just have an irrational desire to find them.

If you want to do science, there has to be a little part of you that just likes something irrationally.

These are the first stars in the universe! We know they have to be there.

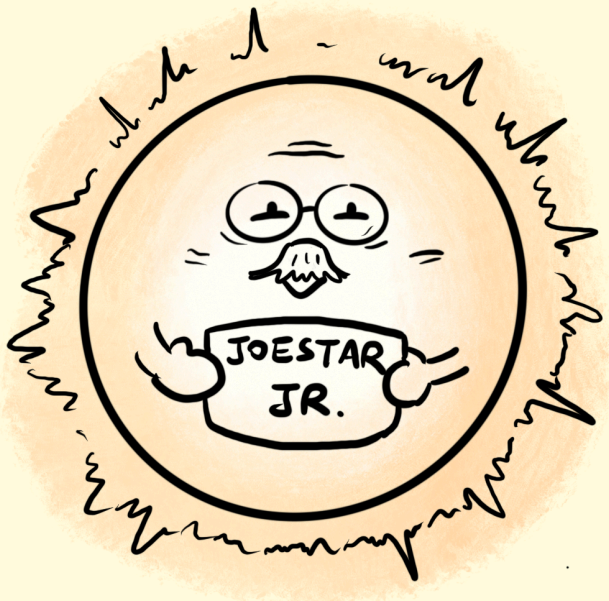
And it's frustrating that we've never seen any [direct] signature of them.





Population III stars are so hard to detect that it may be *impossible* to do so with our current technology.

But their *direct descendants* have been found before,



WELL, HE WAS A BIG STAR...

And they can tell us something about their parent stars.

It was more of these *ancient children* that the undergraduates were determined to find.

From their prepared shortlist of a few dozen suspected metal poor stars,
The plan was to observe about 20 stars every night, spending 10-20 minutes on each star.
Hillary Diane Andales, Graduate Student and Teaching Assistant
Ha Do, Then 3rd Year Undergraduate
Hmm.
Natalie Orrantia, Then 3rd Year Undergraduate
Hmm!
Hillary, what's going on?
The plan did not survive the first night.
ALEX! SELE

A star catalogued as
SDSS J0715-7334
had seized Hillary's attention.

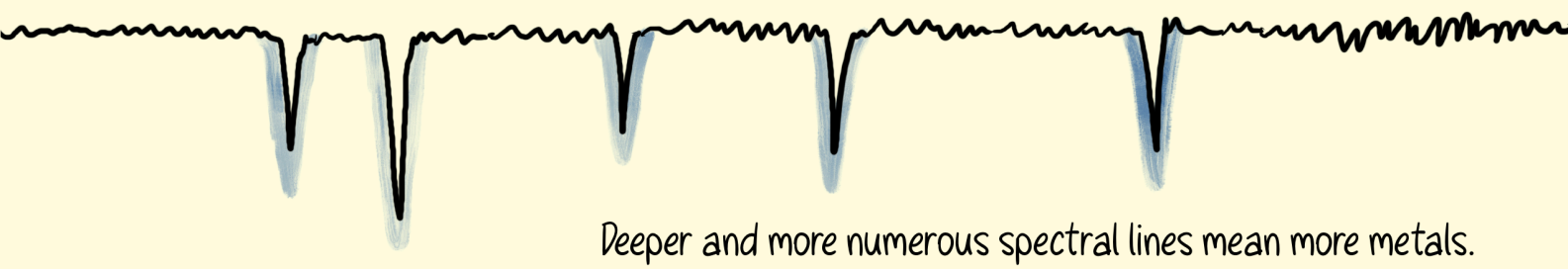
Light from this star had travelled for about 79,000 years to reach Earth.

But what was interesting about the light was what was ***missing.***

Specific colors of light are usually absent from the light we receive from stars,

Because they were absorbed by metals inside the star, before the light ever left.

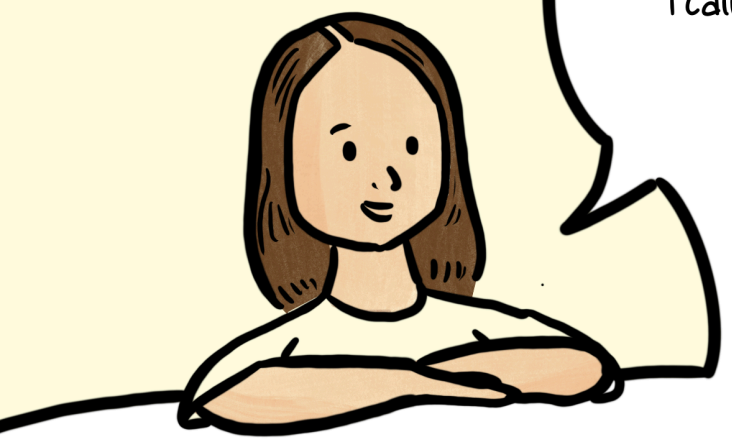
This phenomenon produces characteristic spectral lines in the data.

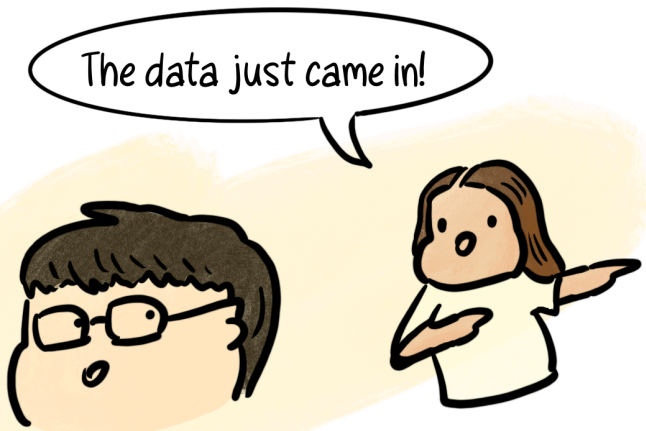


Deeper and more numerous spectral lines mean more metals.

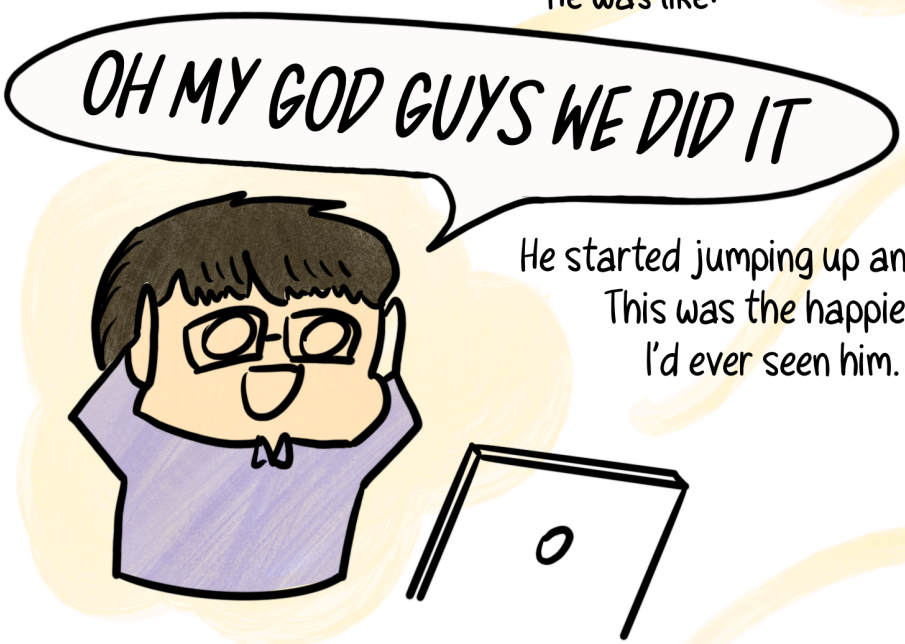
But SDSS J0715-7334 had *nearly no spectral lines at all.*

I called my professor.



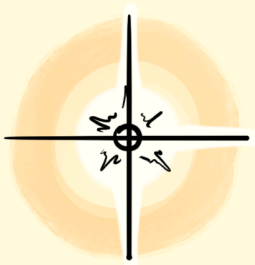


He was like:



He started jumping up and down.
This was the happiest
I'd ever seen him.

The preliminary data showed that SDSS J0715-7334 had an astonishingly low amount of metals.

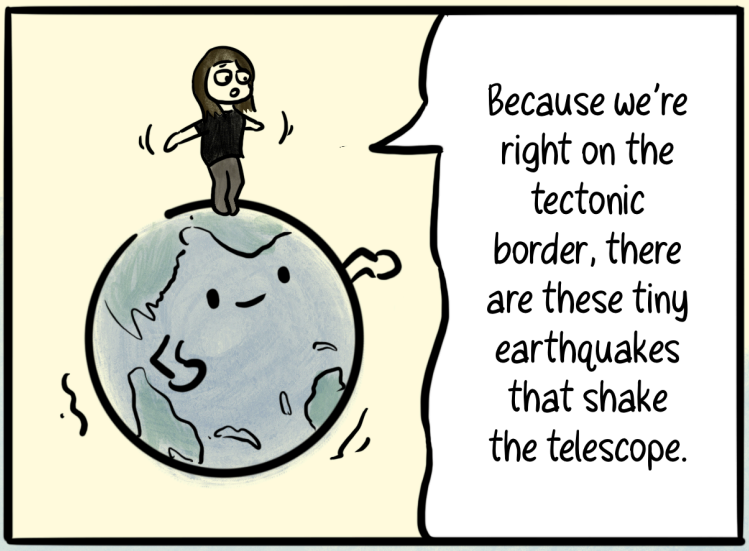


But to be sure, they had to take a longer measurement.

Undergraduates Orrantia and Do spent ***four hours*** the next night observing the star.

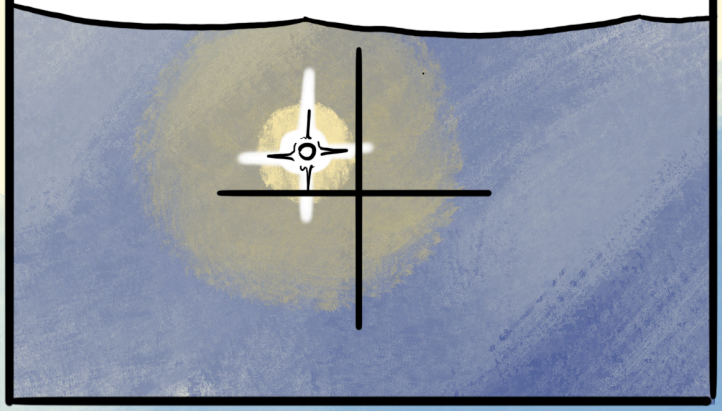
It was a little easier because you're not having to swivel around the telescope to look at new stars.

But it was more ***nerve wracking.***

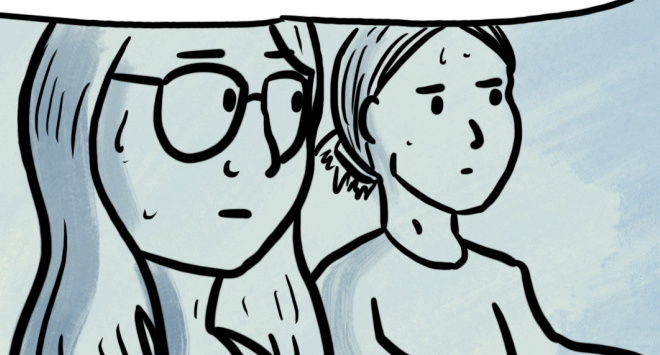


Because we're right on the tectonic border, there are these tiny earthquakes that shake the telescope.

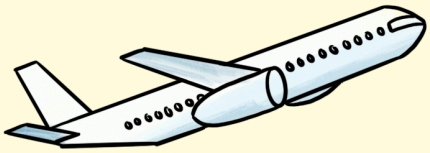
There's a tracking camera for the star. If it moves off target, we get weaker signals.



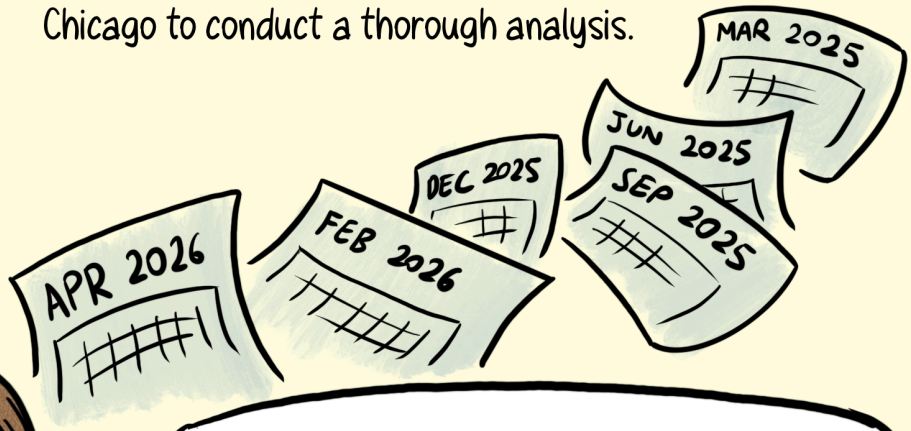
I was making it a point to be very, very quiet to make sure we were focusing on the camera.



We knew whatever we were gonna get from this star could be really important.



With data from 225 minutes of continuous observation in tow, the team returned to Chicago to conduct a thorough analysis.



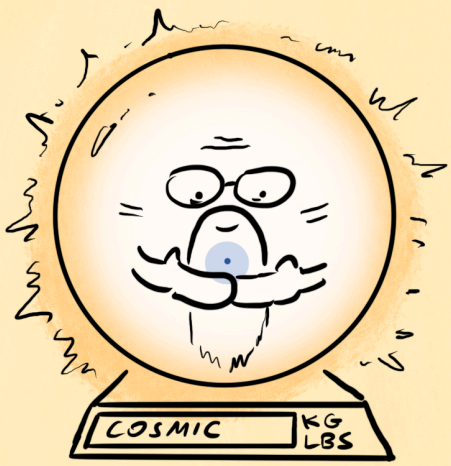
A nearly pristine star from the Large Magallenic Cloud

Published 03 April 2026

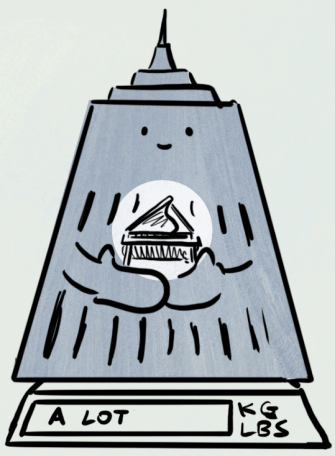
nature astronomy

The claim we ended up making was that this star is the most metal poor star ever discovered by humanity.

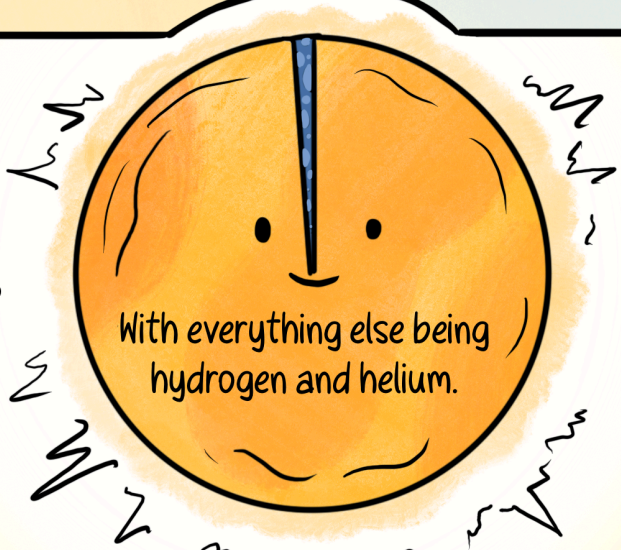
Only about 0.0001% of the star's mass was made up of metals.



That's like the mass of one grand piano compared to the Empire State Building.



For comparison, our much younger sun is 1% metals by mass,



J0715-7334 is more than **10,000** times purer.

For at least one undergraduate, the experience was life-changing.

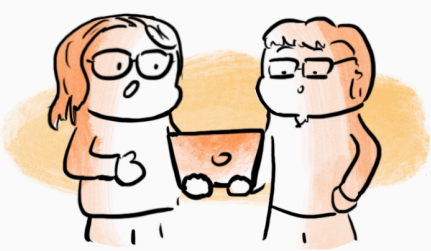
Prior to doing this I sort of thought I wanted to go into engineering.

But I loved going to Chile and actually doing hands-on observing.

Even after getting back, I loved the whole process of computational analysis.

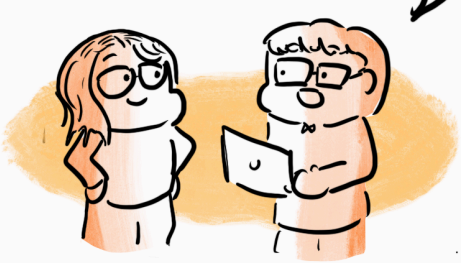
I think I ended up going on some deep Reddit search on what was the best way to do the analysis.

We can use a profile likelihood method for this!



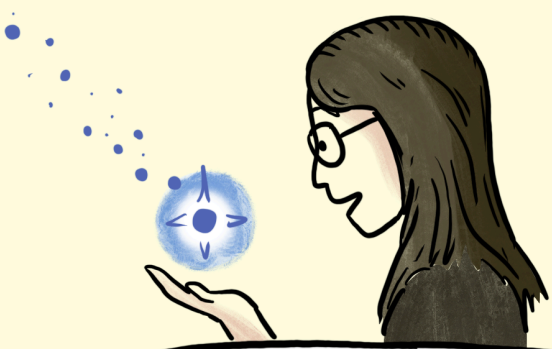
And Alex was like:

I'm really proud that you discovered this.

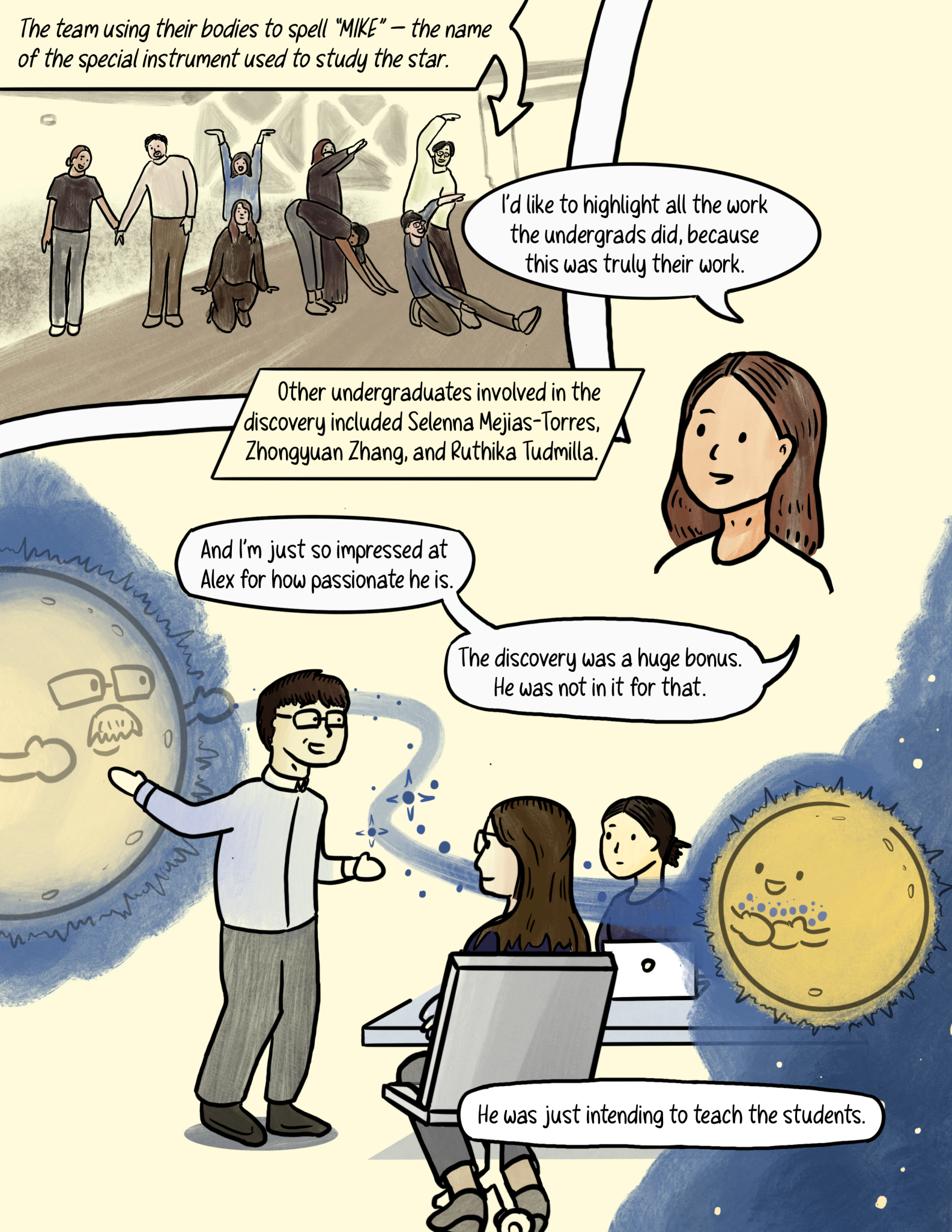


Orrantia originally had had an engineering internship lined up for the summer after the trip to Chile.

She turned it down to pursue research in astrobiology instead, which she hopes to do a PhD in.



I do think this class was the catalyst for my getting back into science.


The team using their bodies to spell "MIKE" – the name of the special instrument used to study the star.
I'd like to highlight all the work the undergrads did, because this was truly their work.
Other undergraduates involved in the discovery included Selenna Mejias-Torres, Zhongyuan Zhang, and Ruthika Tudmilla.
And I'm just so impressed at Alex for how passionate he is.
The discovery was a huge bonus. He was not in it for that.
He was just intending to teach the students.

Reporter's Notes:

To put together this piece of comics journalism, I conducted interviews with three sources, professor Alexander Ji, graduate student Hillary Diane Andales and undergraduate student Natalie Orrantia. Visual references of Las Campanas Observatory (LCO) were obtained from existing public reporting.

In this piece of comics journalism, quotes attributed to sources are denoted through speech bubbles. Illustrations accompanying quotations were my creative decisions, and sources were not asked what kinds of images should accompany each quote.

However, I did solicit my sources for visual information for the purposes of accuracy, such as what the layout of the observation room was like. Undergraduate Calvin O' Brien, not featured in this comic, was also interviewed for information on portraying the rooms of LCO correctly.

This comic was a final project for "Comics for Health and Science Communication", an undergraduate science communication course at the University of Chicago. It was inspired by the work of comics journalist Josh Neufeld.